Appendix - Random Forest

This appendix illustrates the overview of the data processing, analyzing, and modeling of random forest model in this study

Formula

Return rate

$$Return_n = \frac{Close_n}{Close_{n-1}}$$

BIAS

$$BIAS5 = \frac{Close}{MA5} - 1$$

$$BIAS20 = \frac{Close}{MA20} - 1$$

Moving Average Divergence

$$\Delta MA = \frac{MA5}{MA20} - 1$$

Coefficient of Variation

$$CV5 = \frac{Close.std5}{MA5}$$

$$CV20 = \frac{Close.std20}{MA20}$$

Trading Activity

$$\Delta V = \frac{Volume.std5}{Volume.std20} - 1$$

Volatility Momentum

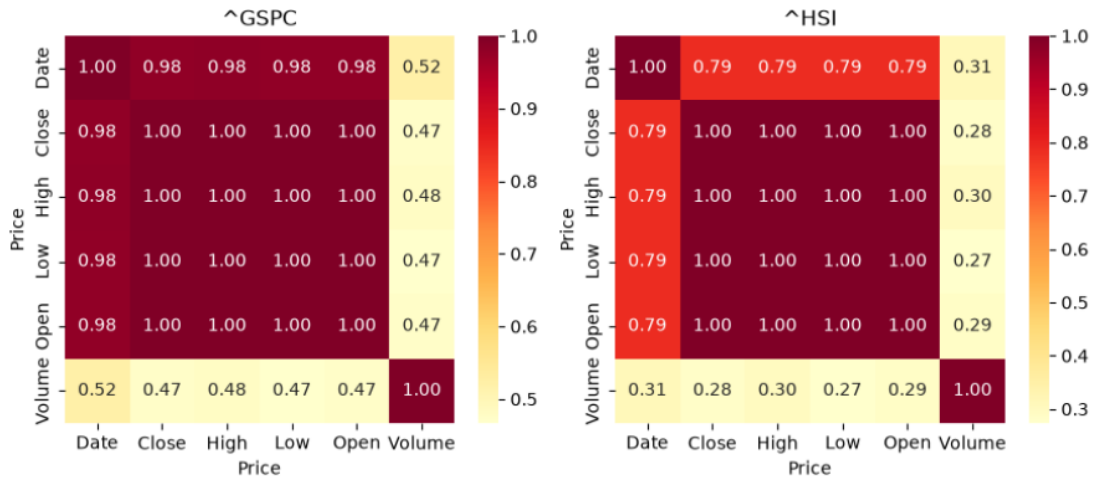
$$\Delta Return = \frac{Return.std5}{Return.std20} - 1$$

Predicted Close

$$pred.close_{n+1} = close_0 \times \prod_0^n pred.return$$

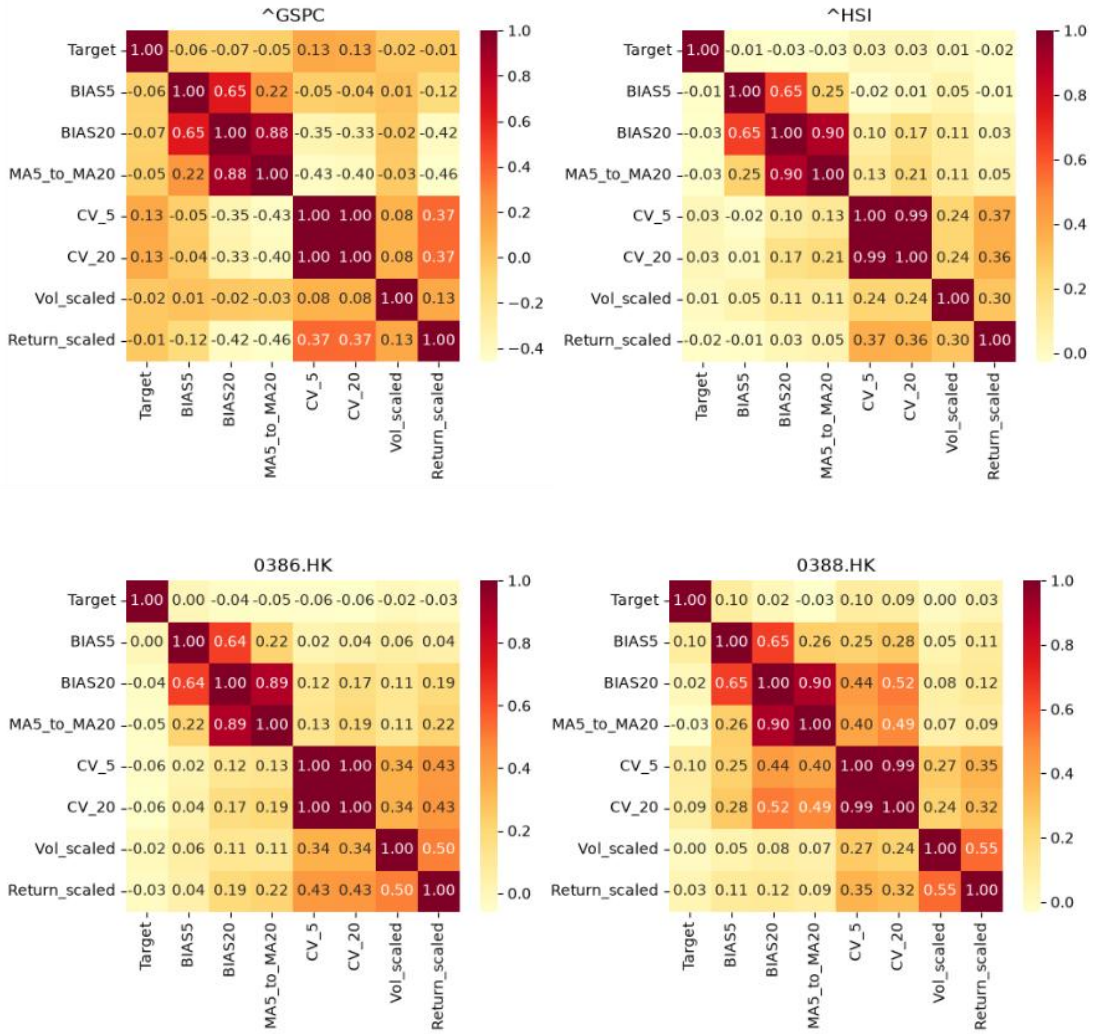
Data Analysis

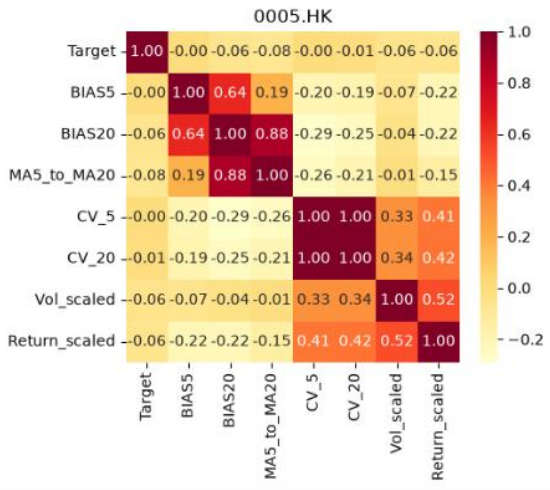
Correlation heatmap between Close and Features



Figures show the correlation heatmap before data processing

Correlation heatmap between Target Return and Features (selected to model prediction)

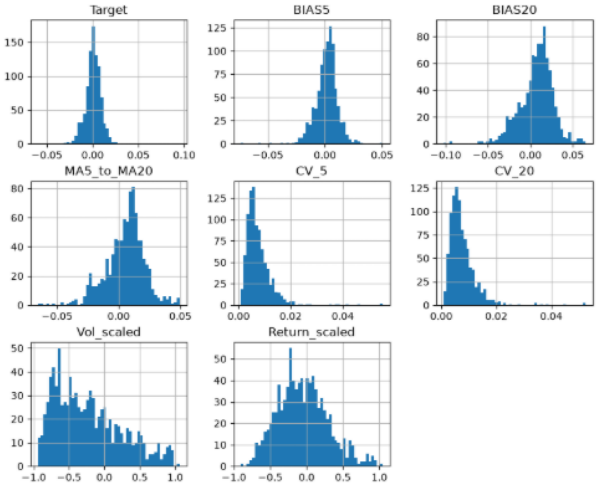




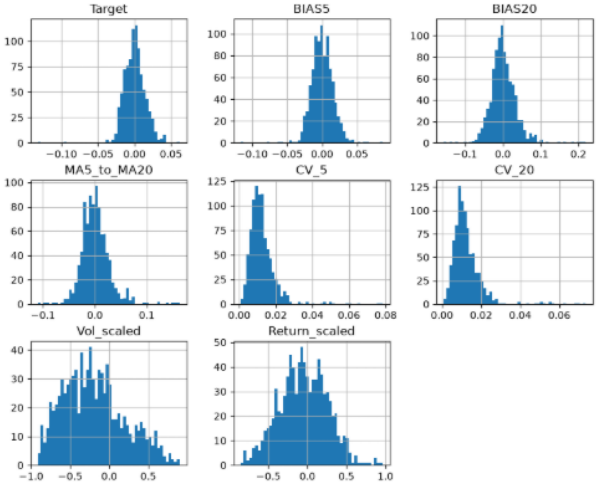
Figures show the correlation heatmap after data processing.

Histogram of the selected features

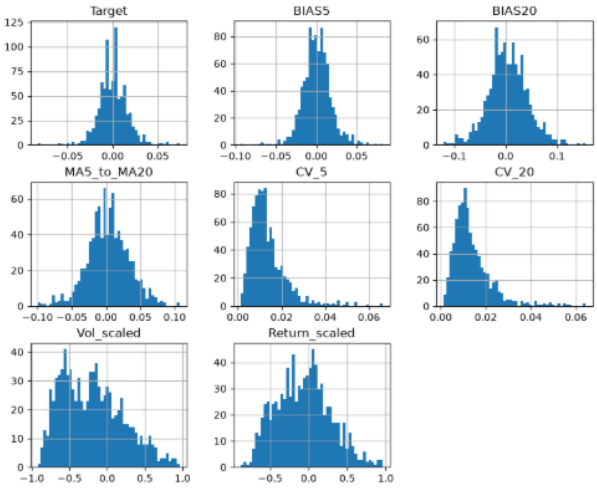
For ^GSPC



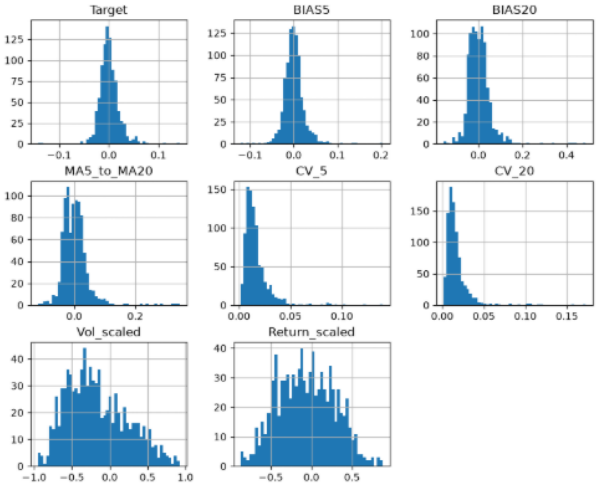
For ^HSI



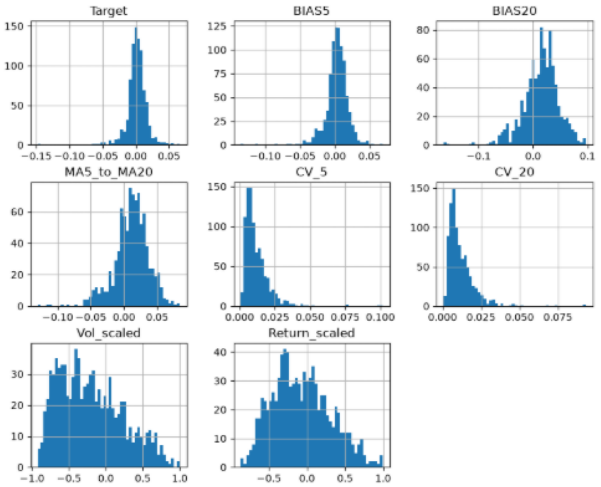
For 0386.HK



For 0388.HK



For 0005.HK

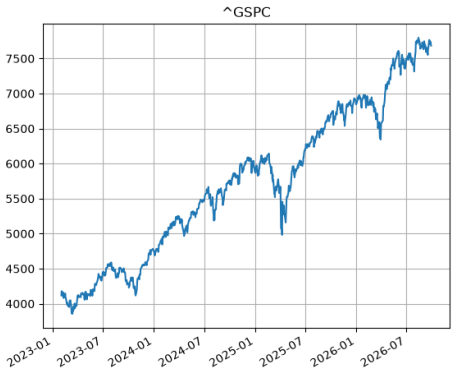


Figures show the distribution of the feature for the model training

Result

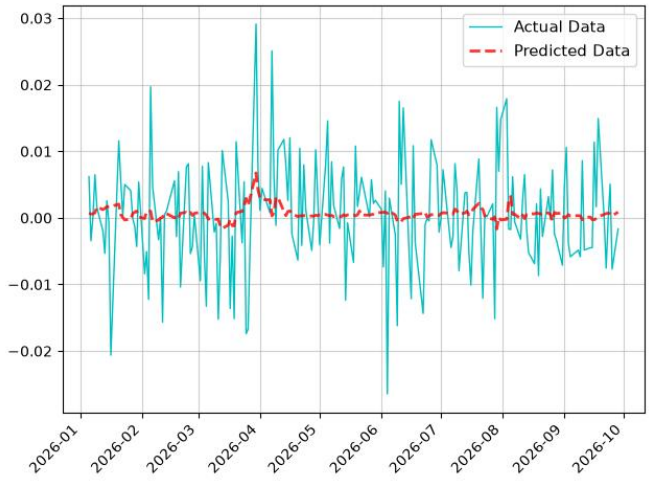
^GSPC

Graph of the Close Price

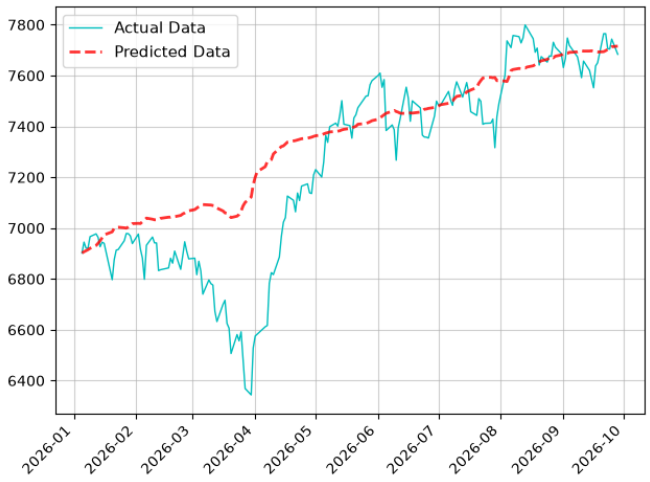


result

Return prediction

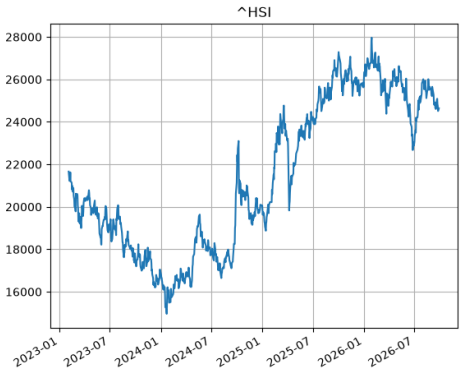


Converted to close prediction



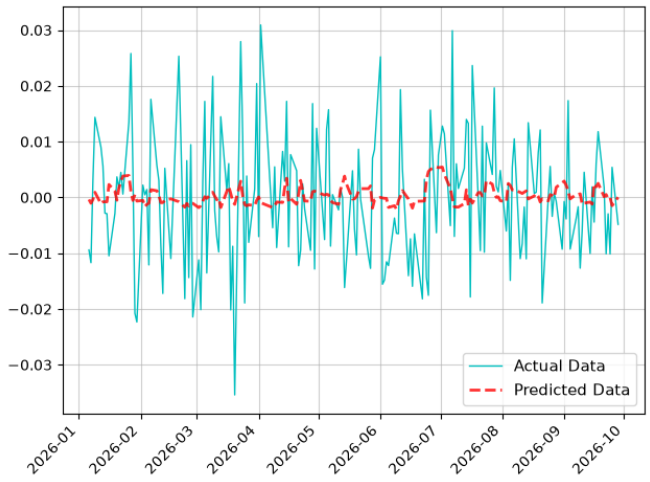
^HSI

Graph of the Close Price

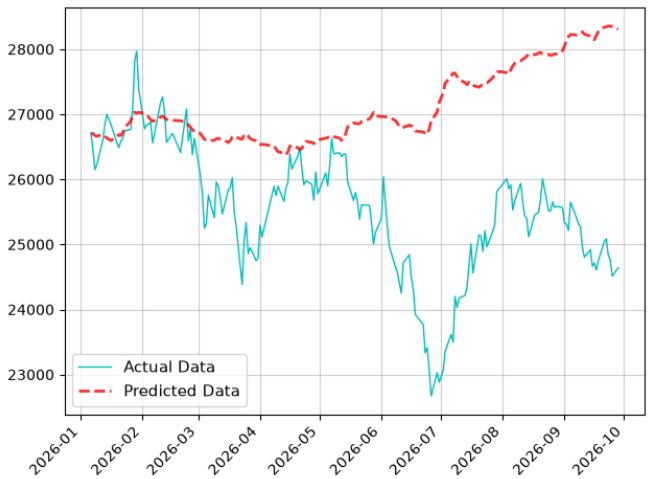


Result

Return prediction

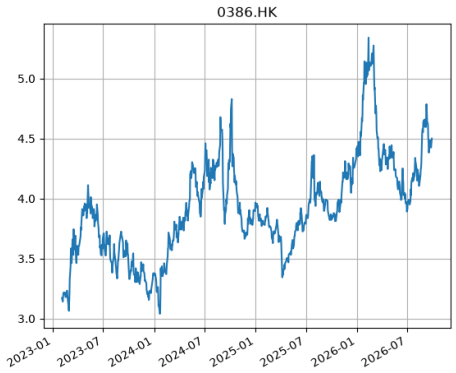


Converted to close prediction



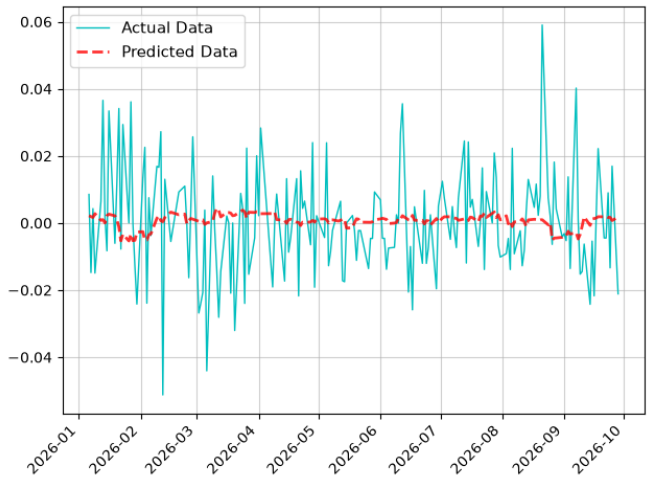
0386.HK

Graph of the Close Price

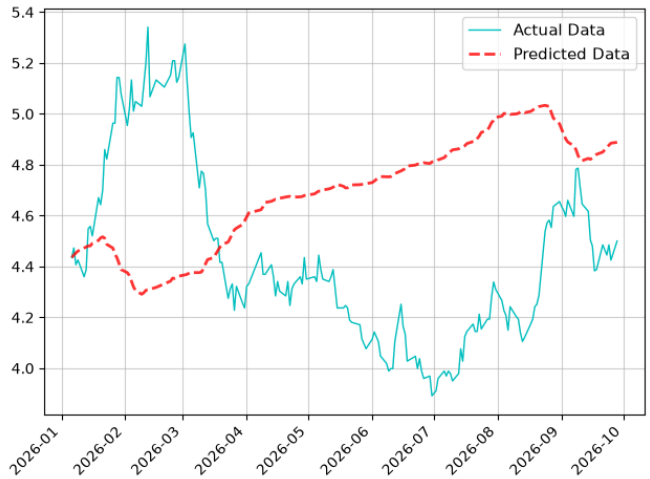


Result

Return prediction



Converted to close prediction



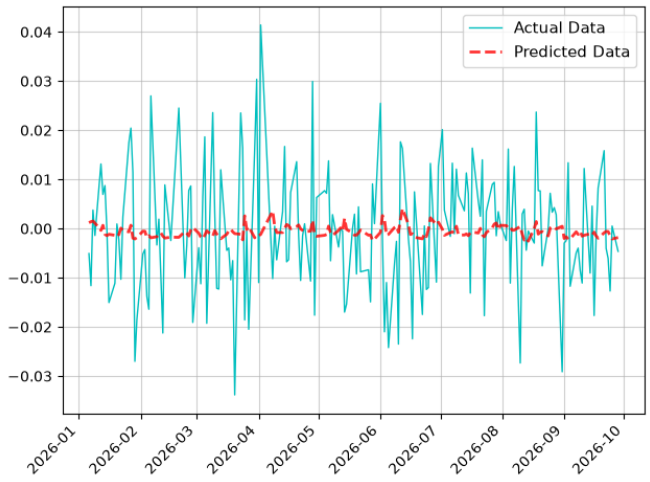
0388.HK

Graph of the Close Price

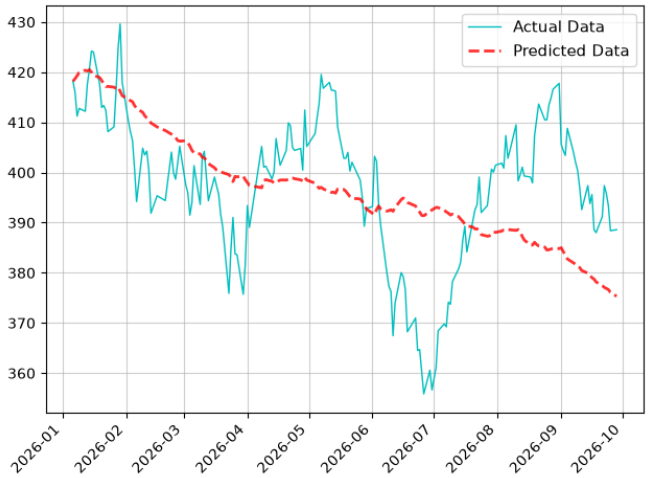


Result

Return prediction

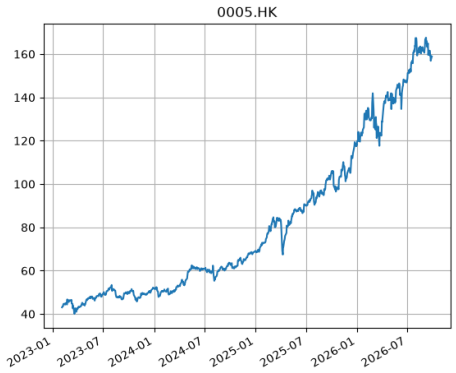


Converted to close prediction



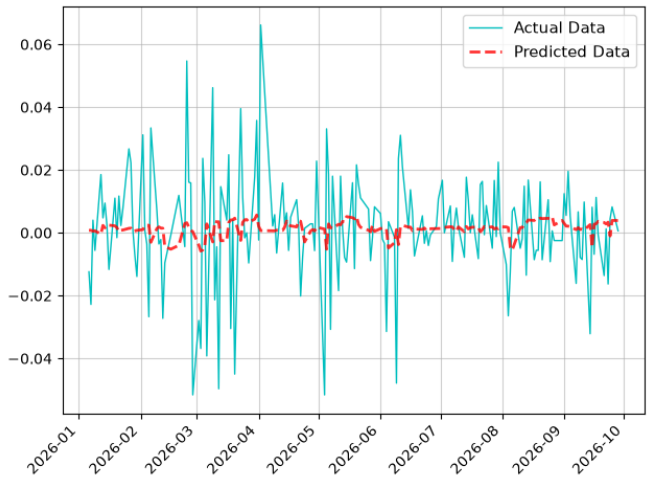
0005.HK

Graph of the Close Price

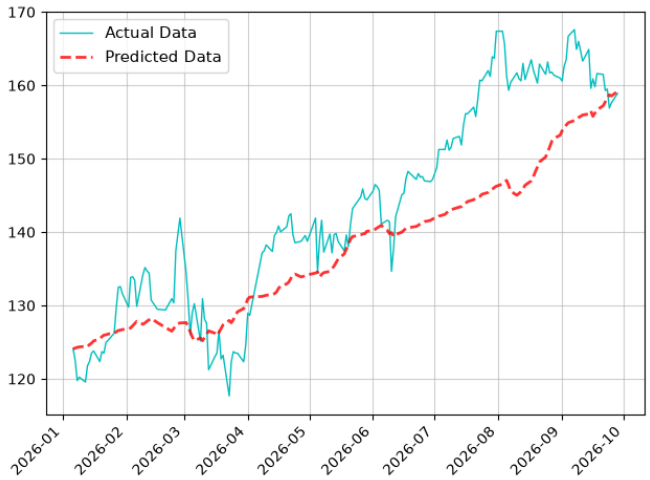


Result

Return prediction



Converted to close prediction



Overall prediction accuracy for Close price prediction

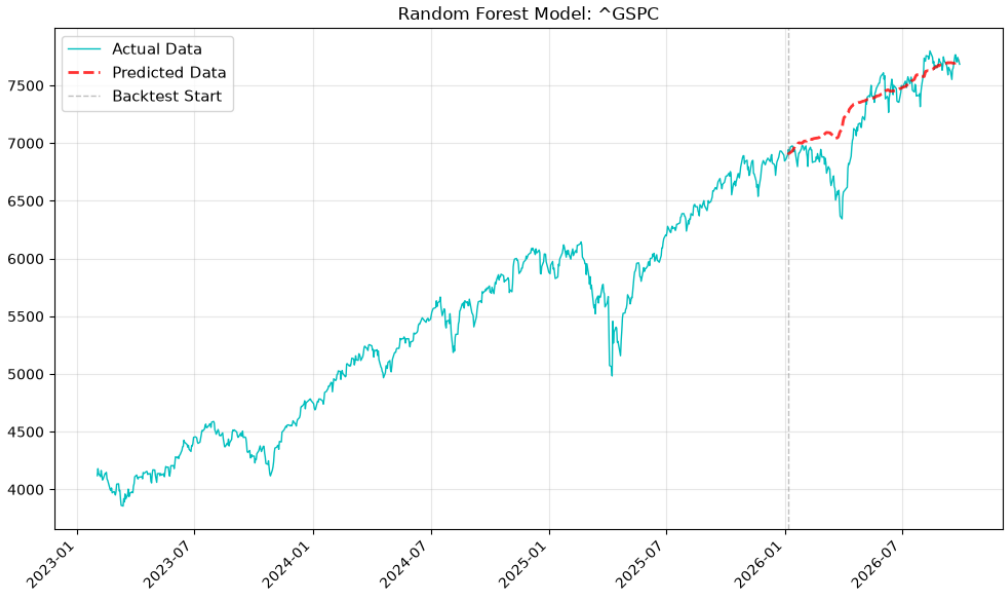
Prediction Accuracy

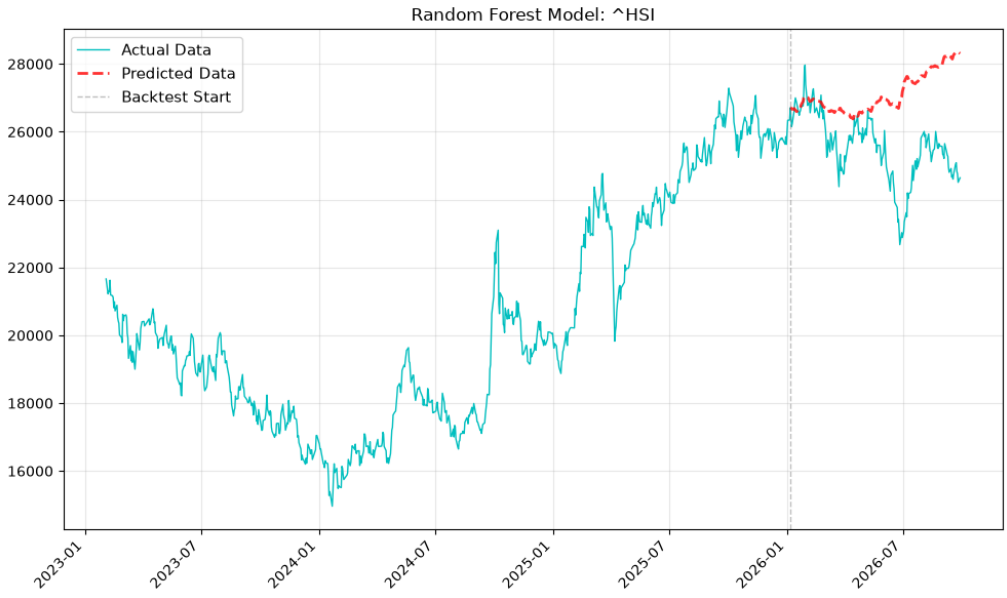
	^GSPC	^HSI	0386.HK	0388.HK	0005.HK
MAE	153.2863	1630.2873	0.5149	12.2786	6.7565
RMSE	223.5761	2036.343	0.5786	14.8926	8.2067
MAPE	2.21%	6.55%	11.77%	3.12%	4.55%
R2 Score	0.6512	-3.3228	-1.7773	-0.0481	0.6661

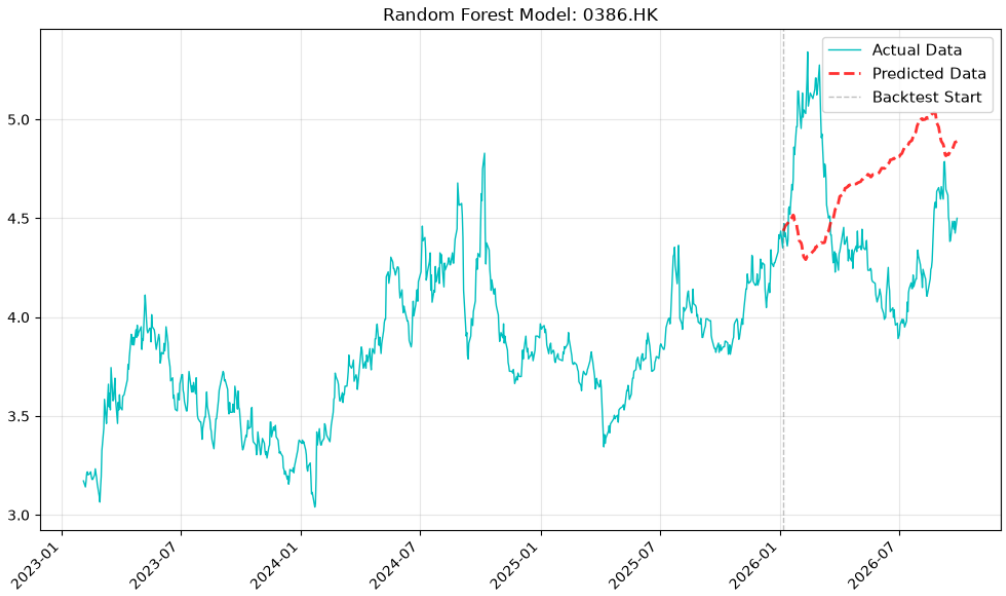
Feature importances

	^GSPC	^HSI	0386	0388	0005
BIAS5	13.20%	4.57%	12.22%	13.87%	35.98%
BIAS20	14.95%	15.41%	15.08%	10.29%	13.11%
MA5/MA20	25.81%	28.23%	33.84%	25.99%	16.24%
CV5	12.42%	16.51%	7.82%	14.10%	7.60%
CV20	11.97%	8.67%	11.01%	10.98%	5.12%
Trading Activity	11.59%	11.98%	9.39%	12.00%	13.13%
Volatility Momentum	10.06%	14.64%	10.63%	12.77%	8.82%

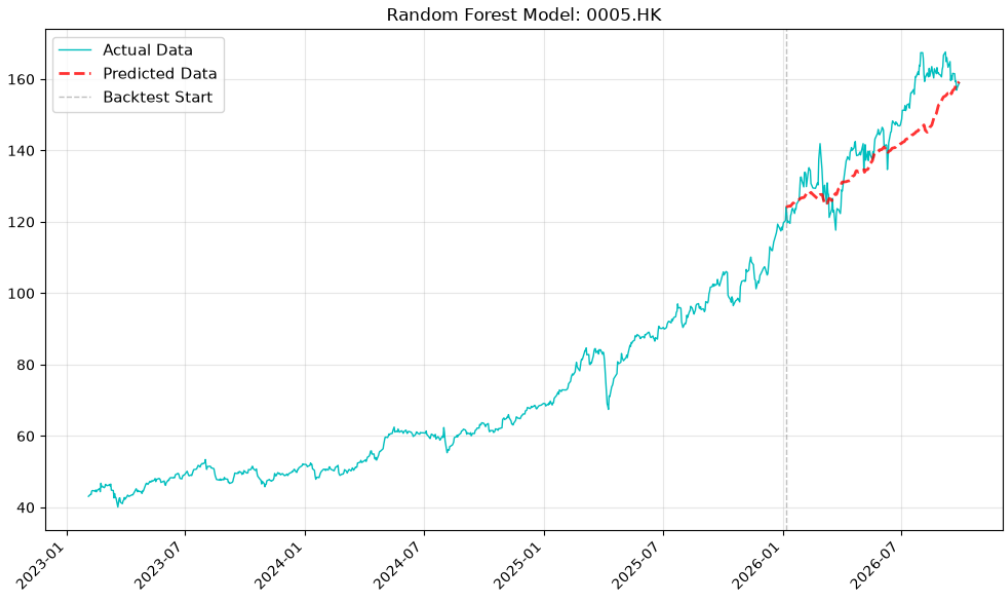
Predicted Curves





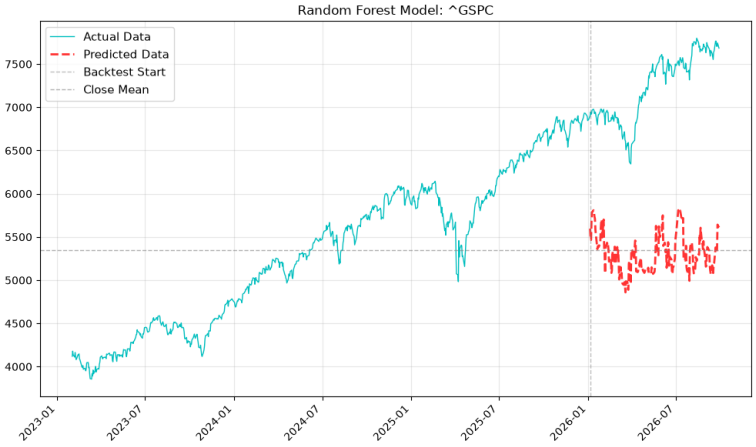

Random Forest Model: 0386.HK
Actual Data
Predicted Data
Backtest Start
5.0
4.5
4.0
3.5
3.0
2023-01
2023-07
2024-01
2024-07
2025-01
2025-07
2026-01
2026-07

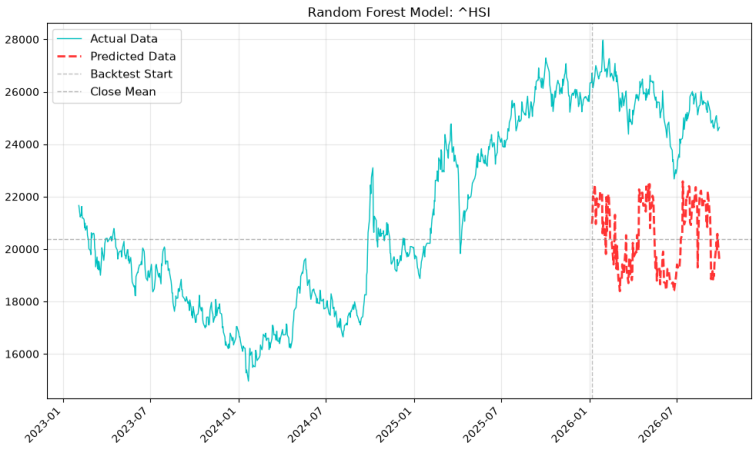

Random Forest Model: 0388.HK
Actual Data
Predicted Data
Backtest Start
450
400
350
300
250
200
2023-01
2023-07
2024-01
2024-07
2025-01
2025-07
2026-01
2026-07

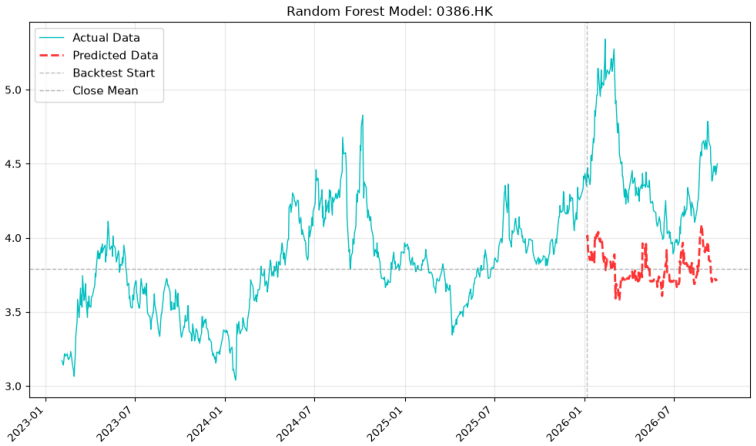


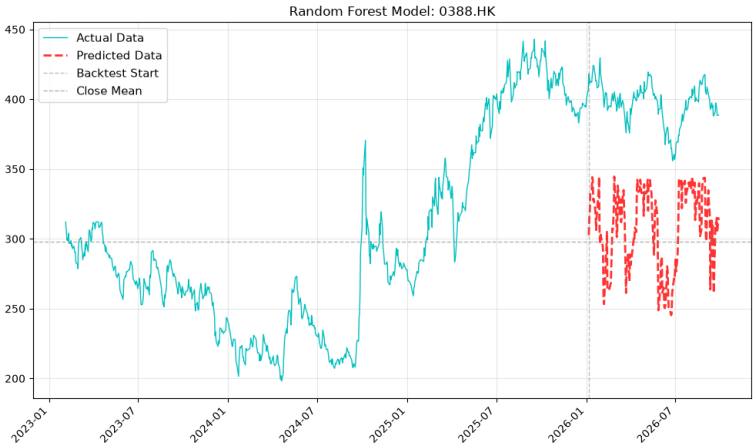
Random Forest failed to predict closing price directly

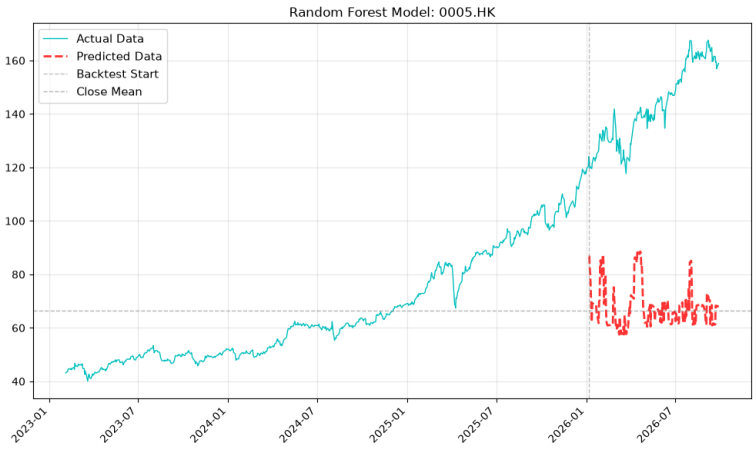
Random Forest cannot extrapolate trends because its predictions are based on averaging the outputs of individual decision trees, which make predictions by partitioning data into regional constants rather than learning mathematical slopes. When a Random Forest model encounters an input value outside the range of its training data, it is unable to project an upward or downward trend and will instead output a flat line capped at the maximum or minimum value seen during training.




Random Forest Model: ^HSI
Actual Data
Predicted Data
Backtest Start
Close Mean
28000
26000
24000
22000
20000
18000
16000
2023-01
2023-07
2024-01
2024-07
2025-01
2025-07
2026-01
2026-07


Random Forest Model: 0386.HK
Actual Data
Predicted Data
Backtest Start
Close Mean
5.0
4.5
4.0
3.5
3.0
2023-01
2023-07
2024-01
2024-07
2025-01
2025-07
2026-01
2026-07


Random Forest Model: 0388.HK
Actual Data
Predicted Data
Backtest Start
Close Mean
450
400
350
300
250
200
2023-01
2023-07
2024-01
2024-07
2025-01
2025-07
2026-01
2026-07


Random Forest Model: 0005.HK
Actual Data
Predicted Data
Backtest Start
Close Mean
160
140
120
100
80
60
40
2023-01
2023-07
2024-01
2024-07
2025-01
2025-07
2026-01
2026-07